18. Consider the data file $mroz$ on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

a. Create two new variables. $MOTHERCOLL$ is a dummy variable equaling one if $MOTHEREDUC > 12$, zero otherwise. Similarly, $FATHERCOLL$ equals one if $FATHEREDUC > 12$ and zero otherwise. What percentage of parents have some college education in this sample?

The percentage of mothers with some college education is 12.1495% , while the percentage of fathers with some college education is 11.6822%.

b. Find the correlations between $EDUC$, $MOTHERCOLL$, and $FATHERCOLL$. Are the magnitudes of these correlations important? Can you make a logical argument why $MOTHERCOLL$ and $FATHERCOLL$ might be better instruments than $MOTHEREDUC$ and $FATHEREDUC$?

	$EDUC$	$MOTHEREDUC$	$FATHERCOLL$
$EDUC$	1.0000	0.3595	0.9850
$MOTHEREDUC$	0.3595	1.0000	0.3546
$FATHERCOLL$	0.9850	0.3546	1.000

The correlations of $EDUC$ with $MOTHERCOLL$ and $FATHERCOLL$ are 0.3595 and 0.3985, respectively, indicating moderate positive relationships. These magnitudes matter because instruments must be correlated with $EDUC$. $MOTHERCOLL$ and $FATHERCOLL$ may be better instruments than parental years of education because they likely contain less measurement error and better capture the effect of having college-educated parents.

c. Estimate the wage equation in Example 10.5 using $MOTHERCOLL$ as the instrumental variable. What is the 95% interval estimate for the coefficient of $EDUC$?

We estimate the following wage equation:

$$ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + \varepsilon$$

where $EDUC$ is treated as endogenous and $MOTHERCOLL$ is used as an instrumental variable.

Using IV estimation, we obtain the coefficient of $EDUC$ as -0.0012. The 95% confidence interval for the coefficient of EDUC is [-0.0012, 0.1533]. Since the confidence interval includes zero, the effect of $EDUC$ on wages is not statistically significant at the 5% level.

d. For the problem in part(c), estimate the first-stage equation. What is the value of the F-test statistic for the hypothesis that $MOTHERCOLL$ has no effect on $EDUC$? Is $MOTHERCOLL$ a strong instrument?

We estimate the first-stage regression:

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 MOTHERCOLL + v$$

To test whether $MOTHERCOLL$ is a relevant instrument, we test: $H_0: \pi_3 = 0$

The F-statistic is 63.563 with a p-value of 1.455×10^{-14}.

Since the F-statistic is much larger than 10 and the p-value is highly significant, we reject the null hypothesis. Therefore, $MOTHERCOLL$ is a strong instrument for $EDUC$.

e. Estimate the wage equation in Example 10.5 using $MOTHERCOLL$ and $FATHERCOLL$ as the instrumental variables. What is the 95% interval estimate for the coefficient of $EDUC$? Is it narrower or wider than the one in part (c)?

We estimate the wage equation using $MOTHERCOLL$ and $FATHERCOLL$ as instrumental variables for $EDUC$:

$$ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + \varepsilon$$

The 95% confidence interval for the coefficient of EDUC is [0.0275, 0.1482].

Compared with part (c), the confidence interval in this part is narrower, suggesting that using both $MOTHERCOLL$ and $FATHERCOLL$ improves estimation precision.

f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of $MOTHERCOLL$ and $FATHERCOLL$. Do these instruments seem adequately strong?

We estimate the first-stage regression:

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 MOTHERCOLL + \pi_4 FATHERCOLL + v$$

To test the joint significance of the instruments, we test:

$$H_0: \pi_3 = \pi_4 = 0$$

The F-statistic is 56.963 with a p-value $< 2.2 \times 10^{-16}$. Since the F-statistic is much larger than 10 and the null hypothesis is strongly rejected, we conclude that $MOTHERCOLL$ and $FATHERCOLL$ are jointly significant. Therefore, the instruments appear to be sufficiently strong.

g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

We test the validity of the overidentifying restrictions using the Sargan test. The null hypothesis is that the surplus instruments are valid, i.e., uncorrelated with the error term.

The Sargan test statistic is 0.238 with a p-value of 0.626. Since the p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, the instruments appear to be valid.